Hanbyul Kim

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EDUCATION

2019 – 2024	**B.S. in Physics** at Hanyang University - Cumulative G.P.A : 3.97/4.0, **Summa Cum Laude** - Early graduation in August of 2024
Sep. 2024 –	**M.S. in Physics (expected Aug. 2027)** at Hanyang University

RESEARCH INTERESTS

My research focuses on emergent quantum phenomena in condensed matter systems, particularly the interplay between symmetry, topology, and magnetism. I am interested in understanding these phenomena through microscopic modeling and symmetry-based approaches.

PUBLICATIONS

[1] **Hanbyul Kim**, Chan Bin Bark, Seik Pak, Gibaik Sim, and Moon Jip Park, "p wave Magnetism and Gate-Tunable Edelstein Response in van der Waals Heterostructures," *Phys. Rev. B* **114**, 074427 (2026); arXiv:2602.11251.

[2] **Hanbyul Kim**, Taewon Yuk, and Sang-Jin Sin, "Topological transitions as percolation of Berry curvature," *J. Phys.: Condens. Matter* **38**, 325501 (2026); arXiv:2407.21098.

[3] Chan Bin Bark, **Hanbyul Kim**, Yoonil Koo, SeungBeom Hong, Seik Pak, Kwang Jun Ahn, Sangmo Cheon, Youngseok Kim, and Moon Jip Park, "An entangling Hubbard quasiparticle collider on a digital quantum processor," arXiv:2610.13946 (under review at Nat. Commun.).

[4] Seik Pak, **Hanbyul Kim**, Chan Bin Bark, Sang-Jin Sin, Jae-yoon Choi, and Moon Jip Park, "η-pairing state in a flatband lattice: Effect of interband coupling on logarithmic scaling of entanglement entropy," *Phys. Rev. B* **114**, 105108 (2026); arXiv:2504.17845.

[5] Chan Bin Bark, **Hanbyul Kim**, Seik Pak, Hong-Guk Min, Sungkyun Ahn, Youngkuk Kim, and Moon Jip Park, "Stacking-dependent topological electronic structures in honeycomb-kagome heterolayers," *npj 2D Mater. Appl.* **9**, 57 (2025); arXiv:2502.14861.

RESEARCH EXPERIENCE

2024 – present **Condensed Matter Research**

- Advisor : Prof. Moon Jip Park
- Topic : p wave Magnetism and Gate-Tunable Edelstein Response in van der Waals Heterostructures

Led this first-author project: derived the Landau free energy and constructed a tight-binding model for a van der Waals heterostructure that stabilizes odd-parity (p wave) magnetic order.

- Topic : Digital Quantum Simulation of Hubbard Quasiparticle Scattering
  Implemented a doublon–holon collider in a Fermi-Hubbard chain on a digital quantum processor, measuring transmission and entanglement generation.
- Topic : Stacking-dependent Topological Electronic Structures in Honeycomb-Kagome Heterolayers
  Calculated the spectrum of the low-angle moiré honeycomb-kagome lattice and discovered a higher-order topological insulator phase under specific symmetry conditions.
- Topic : η-pairing States in Flatband Lattices
  Investigated how interband coupling in flatband lattices modifies the logarithmic entanglement entropy scaling of η-pairing states.

2023 – 2024 **Quantum Geometry Research**

- Advisor : Prof. Sang-Jin Sin
- Topic : Berry-curvature Percolation in Two-dimensional Topological Insulators
  Demonstrated that the percolation of Berry-curvature sign domains determines topological transitions in two-dimensional Chern insulators (first-author publication).
- Topic : Holographic Mean-Field Theory for Strongly Interacting Fermions
  Extended holographic mean-field theory to strongly interacting fermion systems, focusing on symmetry-breaking phenomena and spectral functions; analyzed and classified topological structure using the Hamiltonian method.

2022 – 2023 **Particle Experiment Simulation Research with CERN-CMS**

- Advisor: Prof. Tae-Jeong Kim
- Topic : Searching for the di-Higgs ($t\bar{t}HH$) in the di-lepton state from HL-LHC at $\sqrt{s} = 14$ TeV
  Used Monte Carlo simulations of Phase-II CMS data for background modeling of the $t\bar{t}HH$ process.
  Implemented deep neural networks (DNNs) to enhance the identification of b-tagged jets and classify $t\bar{t}HH$ signals from background events, improving signal detection accuracy.

CONFERENCE

10. 2026	**Korean Physical Society** Fall Meeting 2026 - Presentation in oral session - Topic : Doublon–Holon Scattering on a Quantum Processor
9. 2026	**SCES** International Conference on Strongly Correlated Electron Systems - Presentation in poster session - Topic : p wave Magnetism in van der Waals Heterostructures
03. 2026	**American Physical Society** March Meeting 2026 - Presentation in poster session - Topic : p wave Magnetism in van der Waals Heterostructures
04. 2025	**Korean Physical Society** Spring Meeting 2025 - Presentation in oral session - Topic : Holographic Quantum Geometry Tensor
10. 2024	**Korean Physical Society** Fall Meeting 2024 - Presentation in oral session - Topic : Holographic Quantum Geometry Tensor
05. 2024	**Intercontinental Binodal Workshop**

	on Flat Bands and Higher-order Van Hove Singularities
04. 2024	**Korean Physical Society** Spring Meeting 2024 - Presentation in oral session - Topic : Percolation of the Berry Curvature as Topological Transition
01. 2024	**Workshop on Strongly Correlated Systems and DoS Singularity**
07. 2023	**2023 QUC Summer School on A.I. in High Energy Physics**
06. 2023	**2023 KCMS (Korea CMS collaboration) Workshop**

WORK EXPERIENCE

2024 – present	**Assistant Instructor, General Physics Laboratory** Independently managed all aspects of the laboratory course: designed and delivered instruction, supervised experiments, and grading (2024,2026): General Physics Experiment 2 (course taught entirely in English) (2025,2026): General Physics Experiment 1
2022 – 2024	**Teaching Assistant** (2024): Thermal and Statistical Physics TA (2023): Mathematical Physics TA
2019 – 2024	**Private Tutoring** Tutored mathematics and physics to over 15 students, most of them high school students.

AWARDS AND HONORS

Fall 2024	**Han Ki-su Scholarship** Hanyang University Merit-based support for top-ranked incoming graduate student
2022 – 2023	**Hanyang Academic Excellence Award** recipient of a prize of excellence (4.0/4.0) Only **1%** of the students in the College of Natural Sciences (2022, 2023)
2021 – 2024	**Academic Excellence Award : Merit-based Scholarship** only for **1st, 2nd** place Tuition waiver for the semester at Hanyang University (2021, 2022, 2023)
Spring 2024	**Outstanding Paper Presentation Award** Hanyang University Awarded for excellence in graduation thesis presentation
Fall 2023	**Outstanding Presentation Award** conference at Hanyang University Awarded for exceptional presentation and Academic Activities Scholarship

SKILLS AND QUALIFICATIONS

Programming	Python, Mathematica, ROOT (CERN), Unix/Linux, MATLAB, C, Julia